

AI Assistant

|

아프거나 궁금할 때,
요즘은 AI에게 먼저 묻습니다

AI Assistant

신장 기능이 안 좋은데, 메트
포르민 계속 먹어도 될까?

답은 즉시,
확신에 찬 말투로 돌아옵니다

AI Assistant

신장 기능이 안 좋은데, 메트포르민 계속 먹어도 될까?

신장 기능이 저하된 경우 메트포르민은 중단해야 할 수 있습니다. 정기적으로 검사를 받으시는 것이 좋습니다.

AI Assistant

제2형 당뇨병에 메
트포르민...

신장 기능이 저하된
경우...



그 답, 믿고 따라도 될까요?

일반 LLM

신장 기능이 저하된 경우 메트포르민은 중단
해야 할 수 있습니다.

! 근거를 확인하기 어렵다

의료 판단에는 이런 근거가 필요합니다

A **의료 개념의 관계**
질병 · 약물 · 검사

B **임상 진료지침**
치료 · 검사 · 관리 권고



**질병 · 약물 · 검사는
서로 복잡하게 얽혀 있습니다**

MedicalQA

근거를 함께 보여주는 의료 AI

사용자 질문

A · KNOWLEDGE GRAPH

의료 지식 그래프

질환 · 약물 · 검사의 관계 — SNOMED CT · RxNorm · LOINC

B · CLINICAL GUIDELINE

임상 진료지침

치료 · 검사 · 관리 권고 — WHO · NICE

지식 그래프는 의료 개념의 '연결'을 찾습니다



질환 · 약물 · 검사가 어떻게 이어지는지를 근거로 사용합니다

A 의료 지식 그래프

B 임상 진료지침

두 근거를 함께

LLM

근거를 바탕으로 답변 생성

1

표현 추출

2

개념 연결

3

의도 분류

4

그래프 탐색

5

근거 정리

STEP 1 질문에서 의료 표현 뽑아내기

Mention Detection

SciSpacy

For a patient with COPD experiencing an acute exacerbation, what COPD-related complications should be considered, which medications can be used for treatment, and how should oxygen status be monitored?

의료 자연어 처리 모델이 질문 속 의료 표현만 골라냅니다



STEP 2 표준 의료 개념과 연결하기

Entity Linking

Hybrid Search

Top-3 per Mention

키워드 검색

BM25

의미 검색

Embedding



COPD → **SNOMED CT · 13645005**

Chronic obstructive pulmonary disease

oxygen status → **SNOMED CT · 736015003**

Observable related to oxygenation status

medications → **SNOMED CT · 2533004**

Medication

각 표현마다 상위 3개 후보를 골라 Concept Set을 만듭니다



STEP 3 질문 의도 파악하기

Intent Classification

Keyword + Ontology 기반



DRUG

약물·치료 관련 질문



TEST

검사·측정·모니터링 관련 질문



PATHO

원인·증상·합병증 관련 질문



COMPREHENSIVE

일반·종합 의료 질문





STEP 3 질문 의도 파악하기

Intent Classification

Keyword + Ontology 기반



DRUG

약물·치료 관련 질문



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COMPREHENSIVE

일반·종합 의료 질문



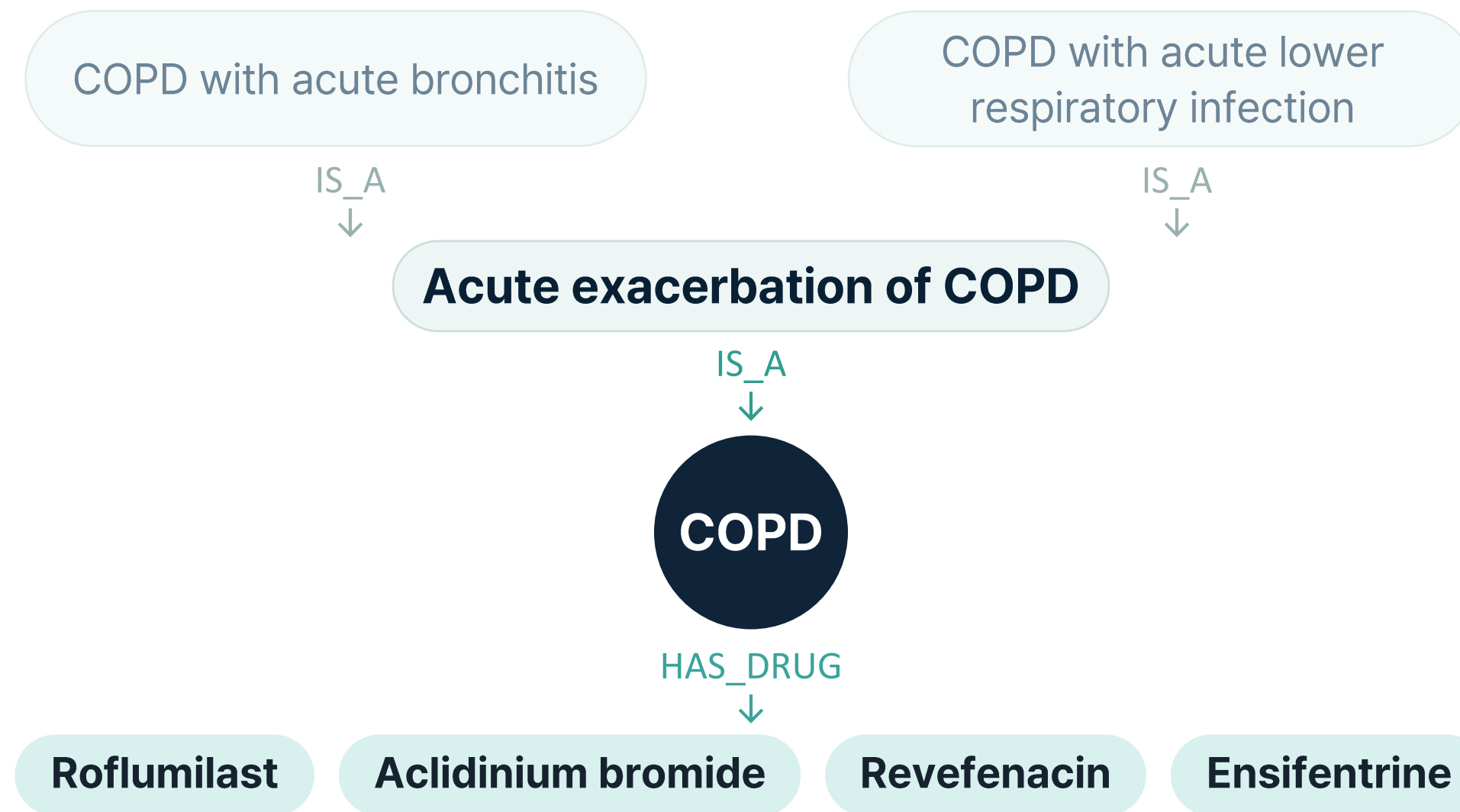
질문의 키워드와 연결된 의료 개념을 바탕으로 하나 이상의 의도를 선택합니다

특정 의도가 탐지되지 않으면 COMPREHENSIVE로 분류

STEP 4 지식 그래프 탐색하기

Neo4j

BFS



질문 의도와 관계별 규칙에 따라 탐색 범위 제한 · 최대 3-Hop

1

표현 추출

2

개념 연결

3

의도 분류

4

그래프 탐색

5

근거 정리

STEP 5 근거 정리하기

Path Filtering

Relation Template

여러 탐색 Path



불필요 · 중복 경로 정리



자연어 Graph Context

Acute exacerbation of COPD is a type of COPD.**Roflumilast may be used to treat COPD.**

별도 경로 진료지침 검색

Qdrant

NICE

Chronic Obstructive Pulmonary Disease

Quality statement 6 — Emergency oxygen during an exacerbation

Oxygen saturation **88–92%**

한편, 진료지침은 질문 임베딩으로 별도 검색됩니다

근거를 만드는 단계

1

표현 추출

SciSpacy

—

2

개념 연결

BM25 + Embedding

—

3

의도 분류

DRUG · TEST · PATHO

—

4

그래프 탐색

Neo4j + BFS

—

5

근거 정리

Graph Context

진료지침 근거 · 별도 경로 Question Embedding → Qdrant → Top-3 Guideline → Guideline Context

Graph Context + Guideline Context →

Gemma

의료 객관식 벤치마크

MedQA

대상 질환 관련 330문항

CKD · COPD · HF · HTN · T2DM

같은 문제 + 같은 선택지

Gemma4-only

MedicalQA

MedQA 정답과 비교

같은 Gemma4:12b, MedicalQA에는 의료 근거 검색을 추가

Gemma4-only

Gemma4:12b

VS

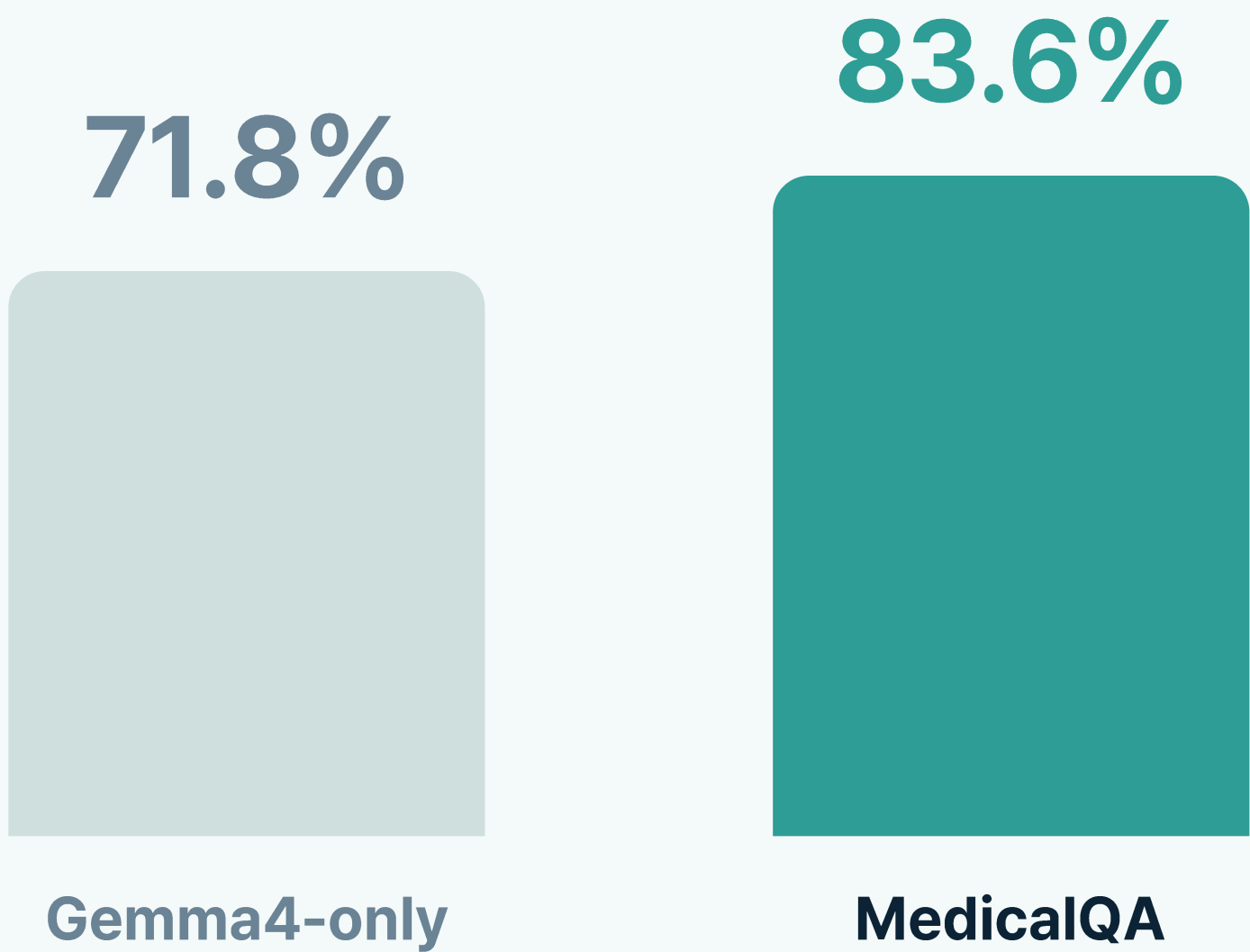
MedicalQA

Gemma4:12b

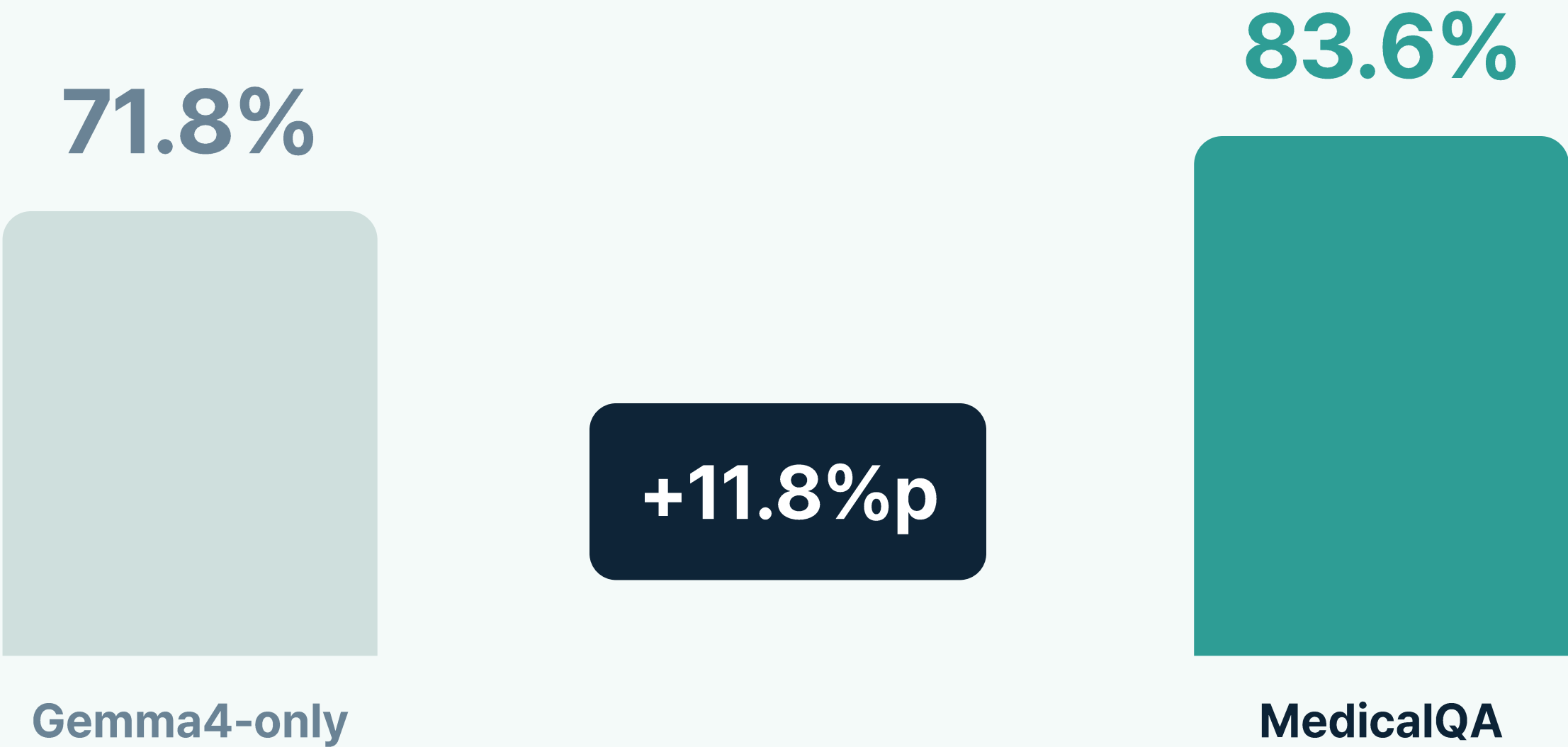
+ 지식 그래프 검색

+ 임상 진료지침 검색

MedQA 정확도 비교



정답률 11.8%포인트 향상



330문항 중 39문항 추가 정답 · MedQA 의료 객관식 벤치마크

01

대표 사례 — ① CKD

폭염 속 훈련 중 쓰러진 환자

폭염 속 훈련 중 쓰러진 환자

환자 상황

폭염 속 훈련 중 쓰러짐

핵심 소견

체온 40.3 °C · 의식 변화 ·
건조한 피부

혼동 단서

아버지의 신장질환 · 청력 · 시각
이상

A 20-year-old woman in the army recruit collapses during an especially hot day at basic training. She was in her normal excellent state of health prior to this event. Two weeks ago, she had an upper respiratory infection (URI) but has since recovered. Her father has chronic kidney disease (CKD), bilateral hearing loss, and vision problems. At the hospital, her temperature is 40.3°C (104.5°F), blood pressure is 85/55 mm Hg, pulse is 105/min, and respiratory rate is 24/min. On physical exam, the patient appears to have altered mental status and her skin is dry, hot, and erythematous. She is complaining of severe bilateral flank pain and generalized myalgia. Catheterization produces 200 mL of tea-colored urine. Urine dipstick is positive for blood, but urinalysis is negative for RBCs or WBCs. Which of the following is most likely responsible for her condition?

A. A mutation of the COL4A5 gene

B. Neisseria meningitides

C. Heat stroke

D. IgA nephropathy

The correct answer is **A**. A mutation of the COL4A5 gene.

Clinical Reasoning:

1. **Clinical Presentation:** The patient presents with acute kidney injury (AKI) characterized by severe flank pain, tea-colored urine (gross hematuria), and a positive dipstick for blood despite no RBCs on microscopy. This "microscopic hematuria" or "ghost" hematuria is highly characteristic of **Alport Syndrome**.

2. **Genetic Link:** The patient's father has chronic kidney disease (CKD) along with hearing loss and vision problems. These are classic hallmarks of Alport Syndrome, which results from mutations in the collagen type IV genes. Specifically:

* **COL4A5** mutations are associated with X-linked Alport syndrome (most common).

* **COL4A3** or **COL4A4** mutations result in autosomal recessive Alport syndrome.

3. **The Trigger:** While the patient is currently experiencing a systemic crisis (high fever, hypotension, and altered mental status), the underlying pathology causing her renal failure and hematuria is genetic. The "tea-colored" urine and the family history of CKD/hearing loss point directly to Alport Syndrome.

4. **Why other options are incorrect:**

* **B (Neisseria meningitidis):** While this could cause sepsis and fever, it does not explain the specific combination of tea-colored urine with a negative microscopic exam or the family history of hearing loss/CKD.

* **C (Heat stroke):** The patient is in a hot environment and has a high temperature;

however, heat stroke alone does not cause "tea-colored" urine without RBCs on microscopy or explain the specific family history provided.

* **D (IgA nephropathy):** This typically presents as gross hematuria following an upper respiratory infection (synpharyngitic hematuria). However, it would show RBCs on microscopic urinalysis and does not involve hearing loss or a specific collagen mutation.

Summary of Alport Syndrome:

Alport syndrome is caused by mutations in the **COL4A3**, **COL4A4**, or **COL4A5** genes, which encode components of type IV collagen in the glomerular and tubule basement membranes. This leads to progressive glomerulonephritis (CKD), hearing loss, and ocular defects. The mutation in **COL4A5** is X-linked, making it the most common form.

✗ 가족력에 과도한 가중치를 두고 소변 소견을 잘못 해석해 Alport syndrome을 선택했다.

A 20-year-old woman in the army recruit collapses during an especially hot day at basic training. She was in her normal excellent state of health prior to this event. Two weeks ago, she had an upper respiratory infection (URI) but has since recovered. Her father has chronic kidney disease (CKD), bilateral hearing loss, and vision problems. At the hospital, her temperature is 40.3°C (104.5°F), blood pressure is 85/55 mm Hg, pulse is 105/min, and respiratory rate is 24/min. On physical exam, the patient appears to have altered mental status and her skin is dry, hot, and erythematous. She is complaining of severe bilateral flank pain and generalized myalgia. Catheterization produces 200 mL of tea-colored urine. Urine dipstick is positive for blood, but urinalysis is negative for RBCs or WBCs. Which of the following is most likely responsible for her condition?

A. A mutation of the COL4A5 gene

B. Neisseria meningitides

C. Heat stroke

D. IgA nephropathy

Based on the clinical presentation provided, the most likely cause of the patient's condition is **heat stroke** (Option C).

Clinical Reasoning:

The patient presents with several hallmark features of heat stroke following intense physical activity in a hot environment:

- Hyperpyrexia:** A core body temperature of 40.3°C (104.5°F) is a primary diagnostic criterion for heat stroke.
- Altered Mental Status:** The patient's "altered mental status" and collapse during training are critical indicators of central nervous system dysfunction caused by extreme heat.
- Physical Findings:** Her skin being "dry, hot, and erythematous" indicates a failure of the body's thermoregulatory mechanisms (anhidrosis), which distinguishes classic heat stroke from exertional heat exhaustion.
- Systemic Involvement:** The presence of severe myalgia and tachycardia are common in cases of significant heat-induced systemic stress.

Analysis of Other Options:

- A. Mutation of the COL4A5 gene:** This is associated with Alport syndrome, which typically presents with hereditary nephritis (hematuria) and hearing loss. While her father has renal issues and hearing loss, this would not cause acute hyperpyrexia or sudden collapse in a healthy 20-year-old during heat exposure.
- B. Neisseria meningitidis:** While meningitis can

cause high fever and altered mental status, it typically presents with signs of infection (like petechiae) rather than the specific "dry/hot" skin associated with environmental heat stroke.

- D. IgA nephropathy:** This is a common cause of glomerulonephritis often following an upper respiratory infection (URI). While she had a URI two weeks ago, IgA nephropathy would present with hematuria and potentially hypertension or renal failure, but it does not explain the acute hyperpyrexia or the specific clinical picture of heat-induced collapse.

Note on Urinalysis: The "tea-colored" urine that is positive for blood on a dipstick but negative for RBCs/WBCs suggests **myoglobinuria**. This occurs when muscle breakdown (rhabdomyolysis) releases myoglobin into the bloodstream, which then filters into the urine. Rhabdomyolysis is a common and severe complication of heat stroke due to extreme muscle exertion in high temperatures.

○ 급성 고체온·의식 변화와 열 노출 상황을 우선 해석해 C. Heat stroke를 선택했다.

× Gemma4-only

가족력에 주목



유전자 신장질환 가능성에 집중



A. COL4A5 mutation 선택 ×

○ MedicalQA

고체온 · 의식 변화 · 열 노출 소견에 주목



Heat stroke 관련 의료 개념·관계 검색



C. Heat stroke 선택 ○

02

대표 사례 — ② HTN

고혈압 약을 먹는 환자에게 생긴 변비와 피로

고혈압 약을 먹는 환자에게 생긴 변비와 피로

환자 상황

45세 남성 · 고혈압 · 고지혈증
약물 치료 중

핵심 소견

변비 · 피로

질문

증상을 유발한 약물의 세포 수준
작용 표적은?

A 45-year-old man comes to his primary-care doctor with his wife for a 4-week history of constipation. The patient reports that his bowel habits have changed from every day to every 3 or 4 days. He also now has to strain to pass stool. On further questioning, his wife has also noticed that he has seemed fatigued, with little interest in going on their usual afternoon walks. Medical history is notable for hypertension and hyperlipidemia, both managed medically. He was last seen for follow-up of these conditions 1 month ago. Physical examination is unremarkable. Which of the following is the cellular target of the medicine that is most likely responsible for this patient's symptoms?

A. Angiotensin-II receptors in the membranes of vascular smooth muscle cells

B. Angiotensin-converting enzyme

C. Na+/Cl- cotransporter in the membranes of epithelial cells in the distal convoluted tubule

D. Na+/K+/2Cl- transporters in the membranes of epithelial cells in the ascending loop of Henle

The patient is presenting with a new onset of constipation and fatigue shortly after starting or continuing treatment for hypertension and hyperlipidemia. In clinical practice, these symptoms are classic side effects associated with calcium channel blockers (CCBs), specifically dihydropyridines like amlodipine or nifedipine. Clinical Reasoning: 1. Constipation: Calcium channel blockers can cause constipation because calcium is required for the contraction of smooth muscle in the gastrointestinal tract; however, more specifically, they are known to cause peripheral edema and systemic effects that can lead to slowed GI motility. More importantly, many patients with hypertension/hyperlipidemia are on CCBs, which are a common culprit for these specific side effects. 2. Fatigue: This is a common non-specific side effect of several antihypertensive medications, including CCBs. 3. The "Distractor" Check: While some other medications (like ACE inhibitors) can cause cough or hyperkalemia, they do not typically present with constipation as a primary side effect. Mechanism of Action: Calcium channel blockers work by inhibiting the influx of calcium ions into smooth muscle and myocardial cells. By blocking these channels, they prevent smooth muscle contraction, leading to vasodilation and decreased peripheral resistance (lowering blood pressure). Option A (Angiotensin-II receptors): This describes ARBs (e.g., Losartan). These do not typically cause constipation. Option B (Angiotensin-converting enzyme): This describes ACE inhibitors. These are associated with a dry cough and angioedema, not primarily constipation. Option C (Na+/Cl- cotransporter): This describes Thiazide diuretics.

✗ 칼슘채널차단제로 잘못 추론한 뒤, 해당 표적이 선택지에 없자 기전이 다른 A를 선택했다.

While they can cause electrolyte imbalances, they are not the primary cause of new-onset constipation in this context. Option D (Na+/K+/2Cl- transporters): This describes Loop diuretics (e.g., Furosemide). These are used for fluid overload and do not typically present with these symptoms. The most likely medication is a Calcium Channel Blocker, which targets the L-type calcium channels in the membranes of vascular smooth muscle cells. Correct Option: A. Angiotensin-II receptors in the membranes of vascular smooth muscle cells (Note: While the mechanism described in A is technically for ARBs, in many standardized medical exams, if a patient presents with these specific symptoms on CCB therapy, and the options provided are slightly modified or intended to test the class, one must look for the most likely pharmacological culprit. However, looking at the standard presentation of this specific vignette, it is designed to point toward Calcium Channel Blockers.) Correction/Refinement: Upon closer inspection of the pharmacology: If the question implies a Calcium Channel Blocker (the drug causing constipation), and Option A describes Angiotensin-II receptors, there may be a slight mismatch in the provided options vs. standard CCB descriptions. However, in many clinical scenarios involving hypertension/hyperlipidemia where "constipation" is the hallmark side effect, Calcium Channel Blockers are the intended answer. The cellular target of Calcium Channel Blockers is the L-type calcium channel. Answer: A (Note: In many versions of this question, Option A describes L-type calcium channels).

A 45-year-old man comes to his primary-care doctor with his wife for a 4-week history of constipation. The patient reports that his bowel habits have changed from every day to every 3 or 4 days. He also now has to strain to pass stool. On further questioning, his wife has also noticed that he has seemed fatigued, with little interest in going on their usual afternoon walks. Medical history is notable for hypertension and hyperlipidemia, both managed medically. He was last seen for follow-up of these conditions 1 month ago. Physical examination is unremarkable. Which of the following is the cellular target of the medicine that is most likely responsible for this patient's symptoms?

A. Angiotensin-II receptors in the membranes of vascular smooth muscle cells

B. Angiotensin-converting enzyme

C. Na⁺/Cl⁻ cotransporter in the membranes of epithelial cells in the distal convoluted tubule

D. Na⁺/K⁺/2Cl⁻ transporters in the membranes of epithelial cells in the ascending loop of Henle

The patient's symptoms of constipation and straining to pass stool, following a period of hypertension management, are highly suggestive of a side effect from a **thiazide diuretic** (such as chlorthalidone or hydrochlorothiazide). Thiazide diuretics are commonly used in the management of systemic arterial hypertension. A known clinical side effect of thiazide diuretics is hypercalcemia and hypokalemia; however, they can also cause significant electrolyte imbalances that may lead to gastrointestinal issues like constipation. More specifically, many patients on these medications experience metabolic changes that contribute to decreased bowel motility. The mechanism of action for **thiazide diuretics** involves the inhibition of the **Na⁺/Cl⁻ cotransporter** in the membranes of epithelial cells in the distal convoluted tubule. By inhibiting this transporter, these drugs decrease the excretion of sodium and chloride, leading to a decrease in urine volume and an increase in the reabsorption of calcium (which is why they are often used in the management of hypercalcemia).

The options provided correspond to different classes of antihypertensive medications:

- * **Option A:** Describes the mechanism of **ACE inhibitors** or **ARBs** (targeting Angiotensin-II receptors), which do not typically cause constipation.
- * **Option B:** Describes the mechanism of **ACE inhibitors** (inhibiting the Angiotensin-converting enzyme).
- * **Option C:** Describes the mechanism of **thiazide diuretics** (inhibiting the Na⁺/Cl⁻ cotransporter in the distal convoluted tubule).
- * **Option D:** Describes the mechanism of **loop**

diuretics (targeting Na⁺/K⁺/2Cl⁻ transporters in the ascending loop of Henle), which are typically reserved for more severe cases of hypertension or heart failure and are less commonly associated with this specific presentation of chronic constipation compared to thiazides. Therefore, the medication most likely responsible for the patient's symptoms is a thiazide diuretic, which targets the Na⁺/Cl⁻ cotransporter in the distal convoluted tubule.

Correct Option: C. Na⁺/Cl⁻ cotransporter in the membranes of epithelial cells in the distal convoluted tubule

○ 증상을 thiazide 계열 약물과 연결하고, 작용 표적인 Na⁺/Cl⁻ cotransporter를 확인해 C를 선택했다.

× Gemma4-only

변비 · 피로를 칼슘채널차단제 부작용으로 판단



추론한 기전이 선택지에 없음



A. Angiotensin-II receptors 선택 ×

○ MedicalQA

고혈압 — thiazide 계열 약물 관계 검색



관련 약물 정보·진료지침 근거 확인



C. Na⁺/Cl⁻ cotransporter 선택 ○

관련 근거는,
특정 단서에 치우친 판단을 보완합니다

01

지식 그래프 + 임상 진료지침 = 근거를 활용한 답변 생성

02

답변과 함께 근거를 직접 확인

03

MedQA 정확도 71.8% → 83.6% (+11.8%p)

감사합니다